PRACTICAL – 2

AIM

To study and understand the different types of network cables used in computer networking.

a) Different Types of Network Cables

Network cables are physical transmission media used to connect computers, networking devices, and other communication equipment. The major types of cables used in networking are:

1. **Unshielded Twisted Pair (UTP) Cable**
2. **Shielded Twisted Pair (STP) Cable**
3. **Shielded and Screened Twisted Pair (SSTP) Cable**
4. **Coaxial Cable**
5. **Fibre Optic Cable**

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1. UN-SHIELDED TWISTED PAIR (UTP) CABLE

UTP cable consists of pairs of insulated copper wires twisted together. It is one of the most commonly used cables in Ethernet networks.

Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
Category 3	10 Mbps	**Advantages:** Low cost, easy to install, small diameter. **Disadvantages:** More susceptible to electromagnetic interference and noise.	10Base-T Ethernet
Category 5	Up to 100 Mbps	Provides better performance than older categories and is easy to install.	Fast Ethernet
Category 5e	Up to 1 Gbps	Improved performance and reduced interference compared with Cat 5.	Fast Ethernet, Gigabit Ethernet
Category 6	Up to 10 Gbps	High speed and improved protection against interference.	Gigabit Ethernet, 10G Ethernet
Category 6a	Up to 10 Gbps	Supports high-speed data transmission over longer distances with reduced interference.	10G Ethernet, data centres

Advantages of UTP

- Cheaper in cost.
- Easy to install and maintain.

- Flexible and lightweight.
- Smaller overall diameter.
- Commonly available.

Disadvantages of UTP

- More prone to electromagnetic interference (EMI).
- More susceptible to external noise than shielded cables.
- Performance can be affected in environments with high electrical interference.

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2. SHIELDED TWISTED PAIR (STP) CABLE

STP cable is similar to UTP but contains additional shielding around the twisted wire pairs. The shielding helps reduce electromagnetic interference and improves signal quality.

Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
Category 6	Up to 10 Gbps	**Advantages:** Shielded, high speed, less susceptible to noise and interference. **Disadvantages:** More expensive and requires greater installation effort.	Gigabit Ethernet, 10G Ethernet
Category 6a	Up to 10 Gbps	Provides high bandwidth and improved protection from interference.	10G Ethernet, data centres

Advantages of STP

- Provides shielding against electromagnetic interference.
- Less susceptible to noise.
- Supports high-speed data transmission.
- Suitable for environments with electrical interference.

Disadvantages of STP

- More expensive than UTP.
- Installation is comparatively difficult.
- Requires proper grounding for effective shielding.
- Slightly thicker and less flexible than UTP.

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3. SHIELDED AND SCREENED TWISTED PAIR (SSTP) CABLE

SSTP cable provides additional shielding around the individual twisted pairs and/or the entire cable. It offers better protection from electromagnetic interference than ordinary UTP cables.

Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
Category 7	Up to 10 Gbps	**Advantages:** Excellent shielding, high bandwidth, low interference. **Disadvantages:** Expensive and requires careful installation.	Gigabit Ethernet, 10G Ethernet

Advantages of SSTP

- Excellent protection from electromagnetic interference.
- Supports high-speed data transmission.
- Suitable for high-performance networking environments.
- Provides better signal quality.

Disadvantages of SSTP

- Higher cost.
- More difficult to install.
- Requires proper grounding and compatible connectors.

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4. COAXIAL CABLE

Coaxial cable consists of a central copper conductor surrounded by insulation, a metallic shield, and an outer protective covering. It is commonly used for television and broadband communication.

Types of Coaxial Cable

Type	Typical Application
RG-59	Television and video connections
RG-6	Cable television and high-speed Internet connections
RG-11	Long-distance cable and broadband applications

Data Transmission

Coaxial cables can support data rates ranging from approximately **10 Mbps to 100 Mbps or higher**, depending on the cable type and network technology.

Advantages

- High bandwidth.
- Good resistance to electromagnetic interference.
- Low signal loss.
- Reliable transmission.
- Versatile and widely used.

Disadvantages

- Bulky compared with twisted-pair cables.
- More difficult to install.
- Limited transmission distance without signal amplification.
- Generally more expensive than basic UTP cable.

Applications

- Cable television networks.
- Broadband Internet connections.
- CCTV and video transmission.
- Older Ethernet networks.

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5. FIBRE OPTIC CABLE

Fibre optic cable transmits data in the form of light pulses through thin strands of glass or plastic fibre. It provides very high speed, bandwidth, and transmission distance.

Types of Fibre Optic Cable

1. **Single-Mode Fibre**
    - Uses a very small core.
    - Suitable for long-distance communication.
    - Provides high bandwidth and low signal loss.
2. **Multi-Mode Fibre**
    - Has a larger core than single-mode fibre.
    - Suitable for shorter-distance communication.
    - Commonly used in LANs and data centres.

Type	Data Transmission	Main Application
Single-Mode Fibre	Very high speed, commonly used for long-distance links	Telecommunications, WANs, long-distance networks
Multi-Mode Fibre	Up to 100 Gbps and beyond depending on the standard	LANs, data centres, campus networks

Advantages

- Very high data transmission speed.
- Very high bandwidth.
- Suitable for long-distance communication.
- Highly secure and difficult to tap.
- Immune to electromagnetic interference.
- Low signal attenuation.
- Lightweight compared with copper cables.

Disadvantages

- Higher installation cost.
- Requires skilled installers.
- Fibre is more delicate than copper cable.
- Special equipment is required for splicing and termination.

Applications

- Internet backbone networks.
- Telecommunications.
- Data centres.
- High-speed LANs.
- Long-distance communication systems.

COMPARISON OF NETWORK CABLES

Cable Type	Transmission Medium	Speed	Interference Protection	Common Applications
UTP	Copper	Up to 10 Gbps depending on category	Low to moderate	LAN, Ethernet
STP	Shielded copper	Up to 10 Gbps	High	Gigabit Ethernet, industrial networks
SSTP	Shielded copper	Up to 10 Gbps	Very high	10G Ethernet, data centres
Coaxial	Copper	10–100 Mbps or higher depending on technology	Good	TV, broadband, CCTV
Fibre Optic	Glass/plastic fibre	Up to 100 Gbps or higher	Immune to EMI	Data centres, Internet backbone, WAN

CONCLUSION

Thus, the different types of network cables such as **UTP, STP, SSTP, coaxial cable, and fibre optic cable** were studied. Each cable has different characteristics, transmission speeds,

advantages, disadvantages, and applications. UTP is economical and widely used for LAN connections, shielded cables are suitable for environments with interference, coaxial cables are commonly used for broadband and television networks, and fibre optic cables are preferred for high-speed and long-distance communication.